Follow-up Milestone Report

Course: Advanced Database Systems
Project Title: Cross-Engine Learned Cost Models for SQL Workloads
Team No.: 7

1. **Project Progress Breakdown:**

ID	Task	Status	Contributor	Notes
1	Project setup & repository structure	Completed	Rinil Parmar	Folder structure, README, requirements.txt, .gitignore created
2	TPC-H data generation (DuckDB)	Completed	Krish Thakkar	SF=0.5, 8 tables generated using DuckDB TPC-H extension
3	Data export to SQLite + indexing	Completed	Hiten Patil	8 tables exported, 20 indexes created, performance PRAGMAs applied
4	Data verification (SQLite vs DuckDB)	Completed	Janitha Salla	Row counts matched across all tables
5	TPC-H query templates (22 queries)	Completed	Lisa Mangnani	All 22 standard TPC-H query templates implemented
6	Parameter randomization for query variants	Completed	Krishal Modi	50 randomized variants per query (~1,100 total queries)
7	Feature extraction engine (25 features)	Completed	Aditi Parmar	25 engine-agnostic structural features extracted from SQL text
8	Benchmarking pipeline (both engines)	Completed	Rinil Parmar	3 runs per query per engine, median runtime recorded
9	Dataset generation & train/val/test split	Completed	Janitha Salla	master_dataset.csv generated, 70/15/15 stratified split
10	Dataset visualization & sanity checks	Completed	Aditi Parmar	Distribution plots, scatter plots, summary statistics generated
11	Model training (4 models × 2 engines)	In Progress	Rinil Parmar, Aditi Parmar	Baseline Linear, Then test for all other Ridge, Random Forest, Gradient Boosting
12	Cross-validation & engine selection accuracy	Pending		K-Fold CV, primary metric: correct engine selection %
13	Baseline comparisons	Pending		Always-SQLite, Always-DuckDB, Join-based heuristic
14	Save best model pair (.joblib)	Pending		model_sqlite.joblib + model_duckdb.joblib
15	Query router / prediction engine	Pending		Input SQL → predicted best engine
16	Evaluation plots & metrics export	Pending		4 figures + evaluation_results.json
17	Failure analysis	Pending		Analyze incorrect engine predictions

18	Documentation & final report	Pending		Full methodology, experiments, and findings

1.2 Task Dependencies & Logical Sequencing

- Data generation in DuckDB → Export + indexing in SQLite
- Verified datasets → Query template generation → Randomized query variants
- Query variants → Feature extraction → Benchmarking pipeline
- Benchmark results → Dataset split → Model training & cross-validation
- Best models → Query router → Evaluation & failure analysis

This sequencing is logical and reflects a well-structured experimental pipeline from data generation to ML-driven system design.

1.3 Progress Summary

- **Core system pipeline is complete**: data generation, verification, feature extraction, benchmarking, and dataset preparation are fully implemented.
- **Experimental foundation is strong:** 1,100+ query variants benchmarked across two engines with reproducible measurements.
- **Advanced ML phase is underway:** multiple regression models are being trained and validated for engine selection accuracy.
- The project demonstrates significant progress and is well beyond a basic prototype stage.

2. GitHub Repository and Version Control:

GitHub link: https://github.com/Rinil-Parmar/cross-engine-learned-cost-model

- The project is fully managed using GitHub for version control.
- All development, experiments, and documentation updates are tracked through commits.

(Note: We don't have enough files to commit for all the group members, as there is 7 members in group. In above table we maintain the history of task.)

Commit ID	Commit Message	Contributor	Contribution Area
fb0aaf0	Initial commit	Rinil Parmar	Repository initialization
665282a	Initial project structure	Rinil Parmar	Project architecture & folder structure
4d1bd25	Add TPC-H dataset generation pipeline with 25 features	Rinil Parmar	Dataset generation & feature extraction engine

a5c04ef	Added script for uploading dataset to Hugging Face	Rinil Parmar	Dataset publishing & external integration
1b1af2a	Final proposal report	Aditi Parmar (Author), Rinil Parmar (Committer)	Research documentation & report integration
3ca972b	Create README.md	Aditi Parmar	Project documentation
	Proposal PPT.	Krishal Modi	Presentation for the project
	Mid Evaluation Report	Krish Thakkar	Progress report (mid milestone)

Version Control:

- Frequent commits tracking meaningful milestones (e.g., TPC-H generation, feature extraction engine, benchmarking pipeline).
- Repository is cleanly structured with modular scripts for data, features, models, and evaluation.
- Commit messages clearly reflect actual work completed.